Teacher Answer Key

Mechanical Oscillations Worksheet (GS + LS) - Ex 1

1. Mechanical Energy

The system is conservative (friction is neglected). Thus, the Mechanical Energy (ME) is conserved during the motion and remains constant, equal to its initial value ME_0 at $t_0 = 0$.

The gravitational potential energy (GPE) is chosen to be zero at the horizontal reference level passing through O ($GPE = 0$).

$$ME = ME_0 = KE_0 + EPE_0$$

$$ME = \frac{1}{2}mv_0^2 + \frac{1}{2}Kx_0^2$$

At time $t_0 = 0$: The solid is at position $x_0 = 0$ (equilibrium) and has velocity $v_0 = 2\ \mathrm{m/s}$.

Substituting the values:

$$ME = \frac{1}{2}(0.2)(2)^2 + \frac{1}{2}(50)(0)^2$$

$$ME = 0.4 + 0$$

Value: $\boxed{ME = 0.4\ \mathrm{J}}$

2. Differential Equation & Nature of Motion

a) Differential Equation:

Since the system is frictionless, Mechanical Energy is conserved:

$$\frac{dME}{dt} = 0$$

$$\frac{d}{dt}\left(\frac{1}{2}mv^2 + \frac{1}{2}Kx^2\right) = 0$$

$$mv\frac{dv}{dt} + Kx\frac{dx}{dt} = 0$$

Knowing that $v = \frac{dx}{dt} = \dot{x}$ and $\frac{dv}{dt} = \ddot{x}$:

$$v(m\ddot{x} + Kx) = 0$$

Since $v \neq 0$ for all t, we have:

$$m\ddot{x} + Kx = 0 \implies \ddot{x} + \frac{K}{m}x = 0$$

Numerical substitution ($m = 0.2, K = 50$):

$$\ddot{x} + \frac{50}{0.2}x = 0$$

Differential Equation: $\boxed{\ddot{x} + 250x = 0}$

b) Nature of Oscillations:

This is a second-order linear differential equation with constant coefficients of the form $\ddot{x} + \omega_0^2 x = 0$, where $\omega_0^2 = \frac{K}{m}$ is a positive constant. Thus, the motion is **Simple Harmonic Motion (SHM)**.

3. Period of Oscillations

The proper period T_0 is given by the formula:

$$T_0 = 2\pi\sqrt{\frac{m}{K}}$$

Substituting the values:

$$T_0 = 2\pi\sqrt{\frac{0.2}{50}} = 2\pi\sqrt{0.004} = 2\pi(0.0632)$$

Period: $\boxed{T_0 \approx 0.40 \text{ s}}$

Representation: This period represents the duration of one complete cycle of the oscillation.

4. Equation of Motion parameters

The solution is of the form $x = X_m \sin(\omega t + \varphi)$. We need to calculate X_m, ω, and φ.

1. Calculation of ω:

Let's verify the differential equation by differentiating the solution x twice:

$\dot{x} = \omega X_m \cos(\omega t + \varphi)$

$\ddot{x} = -\omega^2 X_m \sin(\omega t + \varphi) = -\omega^2 x$

Substitute $\ddot{x}$ and x into the differential equation derived in part 2 ($\ddot{x} + \frac{K}{m}x = 0$):

$$-\omega^2 x + \frac{K}{m}x = 0$$

$$x\left(\frac{K}{m} - \omega^2\right) = 0$$

This relation must hold for any time t (where $x \neq 0$), thus:

$$\frac{K}{m} - \omega^2 = 0 \implies \omega = \sqrt{\frac{K}{m}}$$

$$\omega = \sqrt{\frac{50}{0.2}} = \sqrt{250} \approx 15.8 \text{ rad/s}$$

2. Phase angle (φ):

At $t = 0$, $x(0) = 0$.

$$X_m \sin(\varphi) = 0 \implies \varphi = 0 \text{ or } \pi$$

The velocity expression is $v = \dot{x} = X_m \omega \cos(\omega t + \varphi)$.

At $t = 0$, $v(0) = V_0 = 2 \text{ m/s} > 0$.

$$X_m \omega \cos(\varphi) > 0$$

Since $X_m > 0$ and $\omega > 0$, $\cos(\varphi)$ must be positive. Thus, we select $\varphi = 0$.

Result: $\boxed{\varphi = 0 \text{ rad}}$

3. Amplitude (X_m):

Using the velocity equation at $t = 0$:

$$v(0) = X_m \omega \cos(0) = 2$$

$$X_m = \frac{2}{\omega} = \frac{2}{5\sqrt{10}} \approx 0.126 \text{ m}$$

Result: $\boxed{X_m = 12.6 \text{ cm}}$

Final Equation: $\boxed{x = 0.126 \sin(15.8t)}$ (SI units)

5. Graph of $x = f(t)$

The function is a sine wave starting from 0 and going upwards initially ($v_0 > 0$).

Key points:

- $t = 0, x = 0$
- $t = T_0/4 \approx 0.1s, x = +X_m$ (Max)
- $t = T_0/2 \approx 0.2s, x = 0$
- $t = 3T_0/4 \approx 0.3s, x = -X_m$ (Min)
- $t = T_0 \approx 0.4s, x = 0$